



Laboratorium
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Laporan Akhir

Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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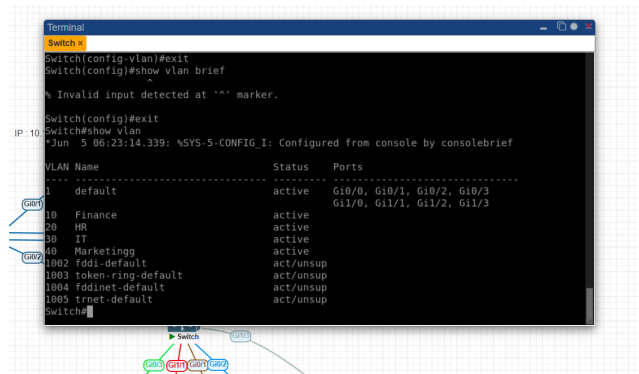
2026

1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Masuk ke mode konfigurasi Cisco Switch menggunakan perintah:

```
enable  
configure terminal
```

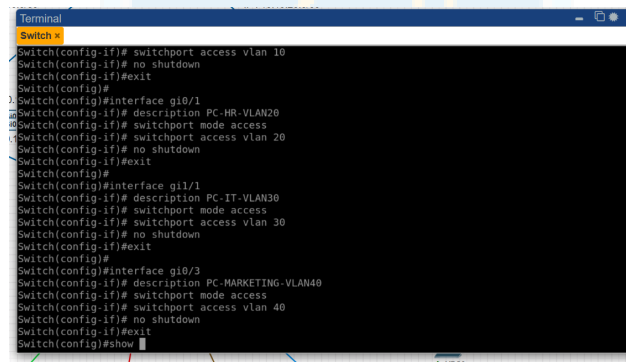


2. Membuat VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing).

3. Melakukan verifikasi VLAN menggunakan perintah:

```
show vlan brief
```

4. Mengkonfigurasi port yang terhubung ke PC sebagai access port sesuai VLAN masing-masing.



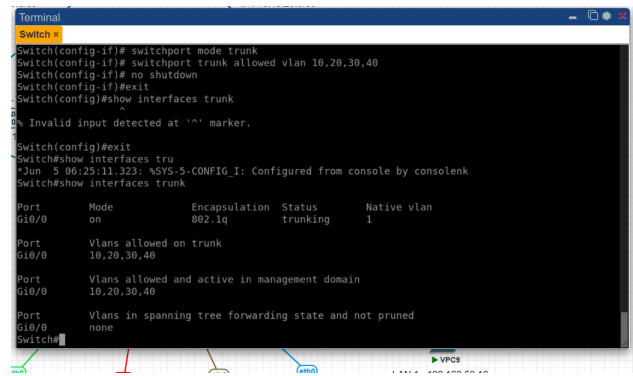
5. Mengkonfigurasi interface yang terhubung ke Cisco Router sebagai trunk port.
6. Mengizinkan VLAN 10, 20, 30, dan 40 melewati trunk menggunakan konfigurasi allowed VLAN.
7. Melakukan verifikasi trunk dengan perintah:

```
show interfaces trunk
```

8. Menyimpan konfigurasi menggunakan perintah:

```
write memory
```

9. Memastikan VLAN dan trunk telah aktif sesuai topologi yang digunakan.



```
Switch#
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30,40
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# show interfaces trunk
% Invalid input detected at '^' marker.
Switch(config)# exit
Switch# show interfaces tru
Jun  5 06:25:11.323: %SYS-5-CONFIG_I: Configured from console by consolenk
Switch# show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
-----
Gi0/0/0   on        802.1q         trunking      1

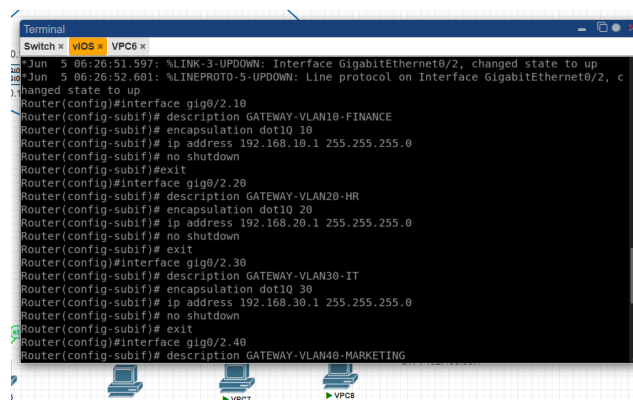
Port      Vlans allowed on trunk
-----
Gi0/0/0   10,20,30,40

Port      Vlans allowed and active in management domain
-----
Gi0/0/0   10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
-----
Gi0/0/0   none
Switch#
```

1.2 Praktikum 2: Konfigurasi Cisco Router

1. Masuk ke mode konfigurasi Cisco Router.
2. Mengkonfigurasi interface yang terhubung ke Cisco Switch sebagai trunk (Router on a Stick).
3. Membuat subinterface VLAN 10, VLAN 20, VLAN 30, dan VLAN 40.
4. Memberikan alamat IP gateway pada masing-masing subinterface.



```
Router#
Router(config)# interface gi0/2.10
Router(config-subif)# description GATEWAY-VLAN10-FINANCE
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)# interface gi0/2.20
Router(config-subif)# description GATEWAY-VLAN20-HR
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)# interface gi0/2.30
Router(config-subif)# description GATEWAY-VLAN30-IT
Router(config-subif)# encapsulation dot1q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)# interface gi0/2.40
Router(config-subif)# description GATEWAY-VLAN40-MARKETING
```

5. Mengkonfigurasi DHCP Server untuk VLAN 10 dan VLAN 20.
6. Mengatur alamat IP statis pada PC VLAN 30 dan VLAN 40.
7. Mengatur PC VLAN 10 dan VLAN 20 agar memperoleh IP secara DHCP.
8. Melakukan verifikasi interface menggunakan:

`show ip interface brief`

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit
Router#
*Jun  5 06:30:53.710: %SYS-5-CONFIG-I: Configured from console by consoleshow
% Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface      IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0      unassigned      YES unset  administratively down  down
GigabitEthernet0/1      unassigned      YES unset  administratively down  down
GigabitEthernet0/2      unassigned      YES unset  up                  up
GigabitEthernet0/2.10   192.168.10.1    YES manual  up                  up
GigabitEthernet0/2.20   192.168.20.1    YES manual  up                  up
GigabitEthernet0/2.30   192.168.30.1    YES manual  up                  up
GigabitEthernet0/2.40   192.168.40.1    YES manual  up                  up
GigabitEthernet0/3      unassigned      YES unset  administratively down  down
GigabitEthernet0/4      unassigned      YES unset  administratively down  down
GigabitEthernet0/5      unassigned      YES unset  administratively down  down
Router#

```

9. Melakukan verifikasi DHCP Pool menggunakan:

show ip dhcp pool

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
10 Router#show ip dhcp pool
Pending event      : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.10.12      192.168.10.1 - 192.168.10.254      1
Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.8d      Jun 06 2026 06:30 AM      Automatic
192.168.20.11    0100.5079.6668.8c      Jun 06 2026 06:30 AM      Automatic
Router#

```

10. Melakukan verifikasi DHCP Binding menggunakan:

show ip dhcp binding

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
10 Router#show ip dhcp binding
Pending event      : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.10.12      192.168.10.1 - 192.168.10.254      1
Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.8d      Jun 06 2026 06:30 AM      Automatic
192.168.20.11    0100.5079.6668.8c      Jun 06 2026 06:30 AM      Automatic
Router#

```

11. Menguji konektivitas dengan melakukan ping ke gateway masing-masing VLAN.

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
0 DDORA IP 192.168.10.11/24 GW 192.168.10.1
3 VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE : 86396, 86400/43200/75600
MAC        : 00:50:79:66:68:8d
I/POR      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.349 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.441 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.866 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=2.238 ms
VPCS>

```



```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x
ODORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.20.1/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86398, 86400/43200/75600
MAC        : 00:50:79:66:68:8c
LPORT      : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=4.099 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=2.122 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.429 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.007 ms
^C
VPCS>

```

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.30.10/24
GATEWAY    : 192.168.30.1
DNS        :
MAC        : 00:50:79:66:68:8b
LPORT      : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.924 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.540 ms
^C
VPCS> [23-]

```

12. Menguji komunikasi antar VLAN untuk memastikan inter-VLAN routing berjalan dengan baik.

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=2.238 ms
^C
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=2.083 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.483 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=3.557 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.623 ms
^C
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=2.163 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=2.921 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=3.802 ms
^C
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.604 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=3.401 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.408 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=3.628 ms
^C
VPCS>

```

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

1. Mengkonfigurasi interface LAN pada Huawei Router dengan jaringan 192.168.50.0/24.
2. Mengkonfigurasi alamat IP pada PC LAN Huawei.

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team.
Press '?' to get help.

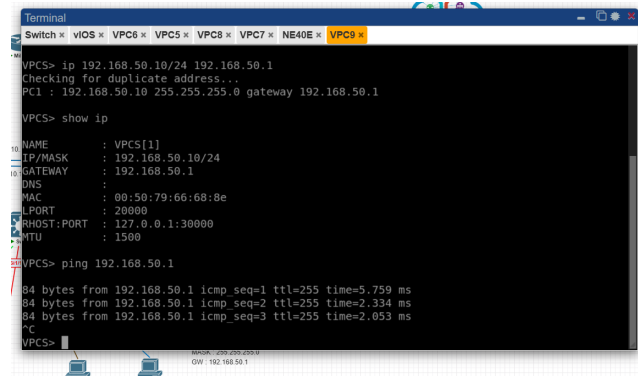
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:8e
LPORT      : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS>

```

3. Melakukan pengujian konektivitas antara PC LAN Huawei dengan gateway Huawei.



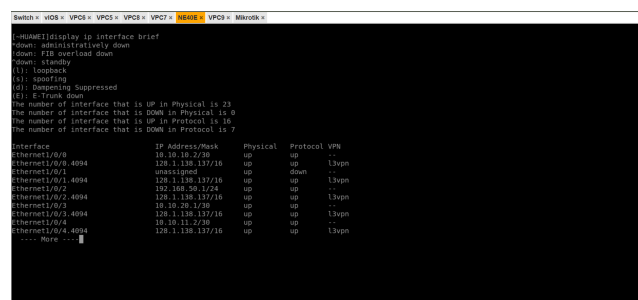
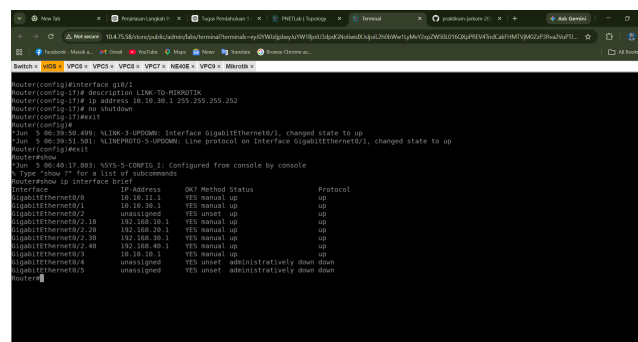
4. Mengkonfigurasi alamat IP pada link Cisco Router dan Huawei Router Link 1 (10.10.10.0/30).

5. Mengkonfigurasi alamat IP pada link Cisco Router dan Huawei Router Link 2 (10.10.11.0/30).

6. Mengkonfigurasi alamat IP pada link Huawei Router dan MikroTik (10.10.20.0/30).

7. Mengkonfigurasi alamat IP pada link Cisco Router dan MikroTik (10.10.30.0/30).

8. Melakukan verifikasi interface pada Cisco Router, Huawei Router, dan MikroTik Router.



```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <

[admin@Mikrotik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@Mikrotik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.10.20.2/30 10.10.20.0 ether1
1 10.10.30.2/30 10.10.30.0 ether2
[admin@Mikrotik] >
```

9. Melakukan pengujian ping antar-router untuk memastikan seluruh link aktif dan dapat saling berkomunikasi.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <

Router(config)#exit
Router#
*Jun  5 06:41:12.127: %SYS-5-CONFIG I: Configured from console by console
Router#display ip interface brief
% Invalid input detected at '^' marker.

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router#
```

10. Memastikan seluruh router dapat melakukan ping ke router tetangga secara langsung.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <

--- 10.10.10.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/2 ms
```

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

[admin@Mikrotik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@Mikrotik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ::: TO-HUAWEI 10.10.20.0 ether1
1 ::: TO-CISCO 10.10.20.0 ether2
[admin@Mikrotik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 2ms
1 10.10.20.1 56 255 1ms
2 10.10.20.1 56 255 6ms
sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=3ms max-rtt=6ms
[admin@Mikrotik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 2ms
2 10.10.30.1 56 255 2ms
sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
[admin@Mikrotik] >

```

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengkonfigurasi OSPF pada Cisco Router dengan Router ID 1.1.1.1.
2. Mengiklankan seluruh network VLAN dan link antar-router ke area OSPF.
3. Mengatur nilai OSPF Cost pada interface Cisco Router sesuai skenario topologi.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Sending updates every 0 seconds
  Inetlink after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum paths 32
  Routing for Networks:
    Routing Information Sources:
      Gateway Distance Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum paths 4
  Routing for Networks:
    10.10.0.0 0.0.0.0 area 0
    10.10.10.0 0.0.0.0 area 0
    10.10.30.0 0.0.0.0 area 0
    102.168.10.0 0.0.0.255 area 0
    102.168.20.0 0.0.0.255 area 0
    102.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway Distance Last Update
  Distance: (default is 110)

Router#

```

4. Mengkonfigurasi OSPF pada Huawei Router dengan Router ID 2.2.2.2.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

[*HUAWEI-Ethernet1/0/3] description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ospf cost 100
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[*HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: None MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00h00m12s
Neighbor Up Time : 2026-06-05 07:01:17
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: None MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m12s
Neighbor Up Time : 2026-06-05 07:01:17
Authentication Sequence: [ 0 ]

[*HUAWEI]

```

5. Mengiklankan seluruh network Huawei ke area OSPF.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]display ospf routing
OSPF Process 1 with Router ID 2.2.2.2
Routing Tables
Routing for Network
Destination Cost Type NextHop AdvRouter Area
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0
Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
--- More ---
```

6. Mengatur nilai OSPF Cost pada interface Huawei Router.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9 Routes : 14
OSPF routing table status : <Active>
Destinations : 5 Routes : 10
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
OSPF 10 110 D 10.10.10.1 Ethernet1/0/4
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
```

7. Mengkonfigurasi OSPF pada MikroTik Router dengan Router ID 3.3.3.3.

8. Mengiklankan network yang terhubung ke MikroTik ke area backbone OSPF.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add net
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add ne
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add n
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network a
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf netwo
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf netw
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf netv
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf net
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf ne
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf n
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf
[admin@mikrotik] > /routing ospf interface add interface=ether1 cost=100
[admin@mikrotik] > /routing ospf interface add interface=ether2 cost=100
[admin@mikrotik] > /routing ospf neighbor print
[admin@mikrotik] > /routing ospf neighbor print
[admin@mikrotik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@mikrotik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=95
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=45
[admin@mikrotik] >
```

9. Mengatur nilai OSPF Cost pada interface MikroTik sebagai jalur cadangan.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routing ospf n
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routing ospf
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routing ospf
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print

[admin@MikroTik] > /routing ospf neighbor print

[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=9s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=4s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
b - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
10.10.30.1
1 Ado 10.10.11.0/30 10.10.20.1 110
10.10.30.1
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.30.1 110
[admin@MikroTik] >

```

10. Melakukan verifikasi OSPF Neighbor pada Cisco Router, Huawei Router, dan MikroTik Router.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x
Routing for Network
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 0)

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 1.1.1.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
10.10.10.0 0.0.0.0 area 0
10.10.11.0 0.0.0.0 area 0
10.10.20.0 0.0.0.0 area 0
192.168.10.0 0.0.0.255 area 0
192.168.20.0 0.0.0.255 area 0
192.168.30.0 0.0.0.255 area 0
192.168.40.0 0.0.0.255 area 0
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 110)

Router#
Jun 5 06:47:14.143: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Jun 5 06:47:14.143: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
Jun 5 06:49:44.472: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 FULL/DR 00:00:36 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:37 10.10.10.2 GigabitEthernet0/3
Router#

```

11. Melakukan verifikasi tabel routing OSPF pada seluruh router.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x
OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 26 sec
Retrans timer interval: 5
Neighbor is up for 00h00m07s
Neighbor Up Time : 2026-06-05 07:01:17
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
--- More ---

```

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
[admin@Mikrotik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@Mikrotik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=9s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=4s
[admin@Mikrotik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS   PREF-SRC  GATEWAY      DISTANCE
0 Ato 10.10.10.0/30          10.10.20.1    110
1 Ato 10.10.11.0/30          10.10.20.1    110
2 Ato 192.168.10.0/24        10.10.30.1    110
3 Ato 192.168.20.0/24        10.10.30.1    110
4 Ato 192.168.30.0/24        10.10.30.1    110
5 Ato 192.168.40.0/24        10.10.30.1    110
6 Ato 192.168.50.0/24        10.10.20.1    110
[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=1m46s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=1m41s
[admin@Mikrotik] >
```

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
IA - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
0 10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:10, GigabitEthernet0/0
1 10.10.20.0/30 [110/110] via 10.10.10.2, 00:04:10, GigabitEthernet0/3
0 192.168.50.0/24 [110/11] via 10.10.11.2, 00:04:10, GigabitEthernet0/0
1 192.168.50.0/24 [110/11] via 10.10.10.2, 00:04:10, GigabitEthernet0/3

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
IA - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
0 10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:30, GigabitEthernet0/0
1 10.10.20.0/30 [110/110] via 10.10.10.2, 00:04:30, GigabitEthernet0/3
0 192.168.50.0/24 [110/11] via 10.10.11.2, 00:04:30, GigabitEthernet0/0
1 192.168.50.0/24 [110/11] via 10.10.10.2, 00:04:30, GigabitEthernet0/3

Router#
```

12. Melakukan pengujian konektivitas end-to-end antara VLAN Cisco dan LAN Huawei menggunakan perintah ping.

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
VPCS>
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.110 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.291 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.684 ms
^C
VPCS>
```

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
VPCS>
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.622 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.673 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=9.393 ms
^C
VPCS>
```

```
Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=5.808 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=5.901 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=4.087 ms
^C
VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=3.972 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=3.828 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=4.909 ms
^C
VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.018 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=5.734 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.692 ms
^C
VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=5.480 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=8.297 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=4.834 ms
^C
VPCS> 
```

13. Memastikan seluruh network telah dipelajari melalui OSPF dan dapat saling berkomunikasi.

```
Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30          10.10.20.1    110
1 ADo 10.10.11.0/30         10.10.30.1    110
2 ADo 192.168.10.0/24        10.10.30.1    110
3 ADo 192.168.20.0/24        10.10.30.1    110
4 ADo 192.168.30.0/24        10.10.30.1    110
5 ADo 192.168.40.0/24        10.10.30.1    110
6 ADo 192.168.50.0/24        10.10.20.1    110
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=lm46s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=lm41s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30          10.10.20.1    110
1 ADo 10.10.11.0/30         10.10.30.1    110
2 ADo 192.168.10.0/24        10.10.30.1    110
3 ADo 192.168.20.0/24        10.10.30.1    110
4 ADo 192.168.30.0/24        10.10.30.1    110
5 ADo 192.168.40.0/24        10.10.30.1    110
6 ADo 192.168.50.0/24        10.10.20.1    110
[admin@MikroTik] > 
```

```
Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

Error: Unrecognized command found at '^' position.
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
.....
public_Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.30.0/30     OSPF   10   110   D  10.10.11.1    Ethernet1/0/4
10.10.10.0/30     OSPF   10   110   D  10.10.10.1    Ethernet1/0/0
192.168.10.0/24   OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24   OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.30.0/24   OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24   OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.50.0/24   OSPF   10   11    D  10.10.11.1    Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.10.0/30     OSPF   10   10    10.10.10.2    Ethernet1/0/0
10.10.11.0/30     OSPF   10   10    10.10.11.2    Ethernet1/0/4
10.10.20.0/30     OSPF   10   100   10.10.20.1    Ethernet1/0/3
192.168.50.0/24   OSPF   10   1     192.168.50.1  Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
```

1.5 Praktikum 5: Pengujian Failover OSPF

1. Melakukan pengecekan routing pada kondisi normal ketika kedua link Cisco-Huawei aktif.
2. Melakukan pengujian jalur menggunakan perintah:

```
trace 192.168.50.10
```


3. Menonaktifkan salah satu link Cisco-Huawei menggunakan perintah shutdown.

```
* 10.10.10.2, from 2.2.2.2, 00:08:00, metric is 11, traffic share count is 1
Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
 0 10.10.10.2 2 msec
 1 10.10.10.2 2 msec
 2 192.168.50.10 6 msec 3 msec 13 msec
```

4. Melakukan pengecekan ulang tabel routing OSPF.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.10.10.0/30 [110/210] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
O    10.10.20.0/30 [110/200] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
C    10.10.30.0/30 is directly connected, GigabitEthernet0/1
L    10.10.30.1/32 is directly connected, GigabitEthernet0/1
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
C    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
```

5. Melakukan trace kembali untuk memastikan lalu lintas tetap melewati jalur utama yang tersedia.

```
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
 0 10.10.10.2 2 msec
 1 10.10.10.2 2 msec
 2 192.168.50.10 6 msec 3 msec 13 msec
```

6. Menonaktifkan link Cisco-Huawei yang kedua sehingga kedua jalur utama terputus.

7. Melakukan pengecekan tabel routing untuk memastikan OSPF melakukan konvergensi.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.10.10.0/30 [110/210] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
O    10.10.20.0/30 [110/200] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
C    10.10.30.0/30 is directly connected, GigabitEthernet0/1
L    10.10.30.1/32 is directly connected, GigabitEthernet0/1
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
C    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
```

8. Melakukan trace kembali dan mengamati perpindahan jalur melalui MikroTik sebagai backup path.

```

192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:17 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:17 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#trace 192.168.50.0
Type escape sequence to abort.
Tracing the route to 192.168.50.0
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.30.2 2 msec 1 msec 1 msec
  2 10.10.20.1 2 msec 2 msec 2 msec
  3 * * *
  4 * * *
  5 * * *
  6 * *

```

9. Mengaktifkan kembali kedua link Cisco-Huawei menggunakan perintah no shutdown.
10. Melakukan verifikasi bahwa jalur routing kembali menggunakan link utama Cisco-Huawei.

```

Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Router#show ip route 192.168.50.10
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:00:08 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:00:08 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:00:08 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip 192.168.50.0
^
% Invalid input detected at '^' marker.
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:00:19 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:00:19 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:00:19 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.10.2 2 msec
  10.10.11.2 3 msec
  10.10.10.2 2 msec
  2 192.168.50.10 8 msec 3 msec 3 msec
Router#

```

2 Analisis Hasil Percobaan

Pada percobaan ini dilakukan konfigurasi FortiGate Jakarta sebagai perangkat edge firewall yang menghubungkan jaringan internal Jakarta dengan jaringan ISP serta jaringan cabang Surabaya melalui GRE Tunnel. Konfigurasi yang dilakukan meliputi pengaturan interface, routing, firewall policy, NAT, GRE Tunnel, dan OSPF. Berdasarkan hasil pengujian, seluruh fungsi utama FortiGate dapat berjalan sesuai dengan tujuan praktikum.

Pada tahap awal, interface FortiGate berhasil dikonfigurasi menggunakan alamat IP yang sesuai dengan tabel addressing. Interface yang terhubung ke Cisco Router Jakarta, MikroTik Router Jakarta, dan MikroTik ISP dapat berstatus *up*. Hal ini menunjukkan bahwa koneksi fisik maupun konfigurasi layer 3 telah berjalan dengan baik. Keberhasilan konfigurasi interface menjadi syarat utama agar proses routing dan pertukaran paket dapat berlangsung.

Selanjutnya dilakukan konfigurasi static route menuju jaringan internal Jakarta dan default route menuju MikroTik ISP. Secara teori, router atau firewall memerlukan informasi routing agar dapat menentukan jalur terbaik menuju jaringan tujuan. Hasil pengujian menunjukkan bahwa FortiGate berhasil mengenali jaringan internal Jakarta serta dapat meneruskan trafik menuju jaringan ISP. Hal ini dibuktikan dengan keberhasilan pengujian ping dari FortiGate menuju alamat publik seperti 8.8.8.8.

Pada proses konfigurasi firewall policy dan NAT diperoleh pemahaman bahwa routing saja tidak cukup untuk menghubungkan jaringan internal ke internet. Walaupun static route telah dikonfigurasi dengan benar, client internal tetap tidak dapat mengakses internet sebelum firewall policy dan NAT

diaktifkan. Setelah policy dibuat dan fitur NAT diaktifkan, paket dari jaringan internal dapat diterjemahkan menjadi alamat IP publik sehingga komunikasi ke internet dapat berjalan. Hasil ini sesuai dengan teori Network Address Translation (NAT), yaitu proses menerjemahkan alamat IP private menjadi alamat IP yang dapat dikenali oleh jaringan publik.

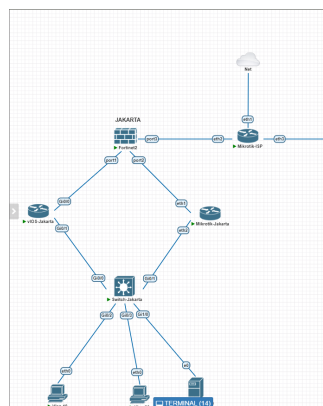
Selama praktikum ditemukan beberapa kendala yang memengaruhi hasil pengujian. Salah satu kendala utama adalah belum adanya konfigurasi NAT pada MikroTik ISP sehingga paket dari jaringan internal tidak dapat mencapai internet. Setelah rule *masquerade* ditambahkan pada MikroTik ISP, konektivitas internet berhasil diperoleh. Kendala lainnya adalah konfigurasi DNS pada Ubuntu Server yang masih menggunakan *localhost resolver* (127.0.0.53), sehingga meskipun koneksi internet telah tersedia, proses *apt update* tetap gagal karena domain tidak dapat diterjemahkan menjadi alamat IP. Setelah DNS diubah menggunakan DNS publik seperti 8.8.8.8 dan 1.1.1.1, proses resolusi nama domain dapat berjalan dengan baik.

Konfigurasi GRE Tunnel dan OSPF juga berhasil diterapkan untuk menghubungkan kantor pusat Jakarta dengan kantor cabang Surabaya. Berdasarkan teori, GRE Tunnel berfungsi sebagai jalur virtual yang memungkinkan dua jaringan berbeda lokasi dapat saling terhubung melalui jaringan publik. OSPF kemudian digunakan untuk mendistribusikan informasi routing secara dinamis melalui tunnel tersebut. Hasil pengujian menunjukkan bahwa neighbor OSPF berhasil terbentuk dan route dari jaringan cabang dapat muncul pada routing table FortiGate Jakarta. Kondisi ini menunjukkan bahwa proses pertukaran informasi routing berlangsung dengan baik sesuai prinsip kerja OSPF.

Secara keseluruhan, percobaan berhasil mencapai tujuan yang telah ditetapkan. FortiGate Jakarta berhasil berfungsi sebagai gateway, firewall, NAT device, endpoint GRE Tunnel, serta router OSPF. Jaringan internal dapat terhubung ke internet dan komunikasi antar-site dapat dilakukan melalui tunnel yang telah dibuat. Kendala yang muncul selama proses konfigurasi sebagian besar disebabkan oleh kesalahan konfigurasi routing, NAT, dan DNS, bukan karena kegagalan perangkat. Oleh karena itu, dapat disimpulkan bahwa pemahaman mengenai hubungan antara routing, firewall policy, NAT, DNS, GRE Tunnel, dan OSPF sangat penting untuk membangun jaringan enterprise yang dapat beroperasi dengan baik.

3 Hasil Tugas Modul

1. Tugas modul 1



Gambar 1: 1.1



2. Tugas modul 2



```

Switch-Jakarta x Switch-Sura... x VIOS-Jakarta x Mikrotik-Jak... x Ubuntu-Vian... x
Mikrotik-Sur... x Fortinet3 x Vlan 30 x Vlan 40 x Tinycore-Via... x

!
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.2 255.255.255.0
 ip helper-address 192.168.60.10
 vrrp 10 ip 192.168.10.1
 vrrp 10 priority 120
end

Router#show running-config interface gi0/1.20
Building configuration...

Current configuration : 181 bytes
!
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.2 255.255.255.0
 ip helper-address 192.168.60.10
 vrrp 20 ip 192.168.20.1
 vrrp 20 priority 90
end

Router#show running-config interface gi0/1.60
Building configuration...

Current configuration : 149 bytes
!
interface GigabitEthernet0/1.60
 encapsulation dot1Q 60
 ip address 192.168.60.2 255.255.255.0
 vrrp 60 ip 192.168.60.1
 vrrp 60 priority 120
end

```

Gambar 6: 2.3

```

Router#ping 10.10.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router#

```

Gambar 7: 2.4

3. Tugas modul 3

```

Router#show ip interface vrrp print
Flags: X - disabled, I - invalid, D - dynamic
# Name INTF... MAC-ADDRESS VRRP PRI INTERVAL V V3...
0 0 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4
2 2 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4
4 4 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4
0 0 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4

```

Gambar 8: 3.1

```

Router#show ip interface vrrp print
Flags: X - disabled, I - invalid, R - running, M - master, D - backup
# Name INTF... MAC-ADDRESS VRRP PRI INTERVAL V V3...
0 0 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4
2 2 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4
4 4 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4
0 0 vrrp20 vlan... 00:00:50:00:01:04 20 120 1s 2 ip4

```

Gambar 9: 3.2

```

Router#show ip dhcp relay print
Flags: X - disabled, I - invalid
# Name INTF... DHCP-SERVER LOCAL-ADDRESS
0 X1 relay vlan10-finance 192.168.60.10 192.168.10.3
1 X1 relay vlan20-11 192.168.60.10 192.168.20.3

```

Gambar 10: 3.3

```

Router#show ip route print
Flags: X - disabled, A - active, D - dynamic
# Name INTF... MAC-ADDRESS VRRP PRI INTERVAL V V3...
0 A S 0.0.0.0/0 10.10.101.2 10.10.101.1 0
2 ADC 192.168.10.0/24 192.168.10.3 vlan10-finance 0
3 ADC 192.168.20.0/24 192.168.20.3 vlan20-11 0
4 ADC 192.168.60.0/24 192.168.60.1 vrrp20 0
5 ADC 192.168.60.0/24 192.168.60.3 vrrp20-server 0

```

Gambar 11: 3.4

Gambar 12: 3.5

[illegible]

Gambar 13: 4.1

```
ip default-gateway 192.168.0.25
hostname "R1-Router"
interface eth0
  ip address 192.168.0.25 255.255.255.0
  no shutdown
end
```

Gambar 14: 4.2

```
root@hcn:~# sudo cat /etc/dhcp/dhcpd.conf
authoritative;

default-lease-time 600;
max-lease-time 7200;

option domain-name-servers 8.8.8.8, 1.1.1.1;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.100 192.168.10.200;
    option routers 192.168.10.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.10.255;
}

subnet 192.168.20.0 netmask 255.255.255.0 {
    range 192.168.20.100 192.168.20.200;
    option routers 192.168.20.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.20.255;
}

subnet 192.168.60.0 netmask 255.255.255.0 {
    option routers 192.168.60.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.60.255;
}

root@hcn:~#
```

Gambar 15: 4.3

```
root@kvm:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=110 time=25.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=110 time=25.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=110 time=26.0 ms
^C
-- 8.8.8.8 ping statistics --
3 packets transmitted, 3 received, 0% packet loss, time 2000ms
rtt min/avg/max/mdev = 25.579/25.812/26.047/0.191 ms
root@kvm:~#
```

Gambar 16: 4.4

[illegible]

Gambar 17: 5.1

Gambar 19: 5.3

Gambar 18: 5.2

Gambar 20: 5.4

Gambar 21: 5.5

Gambar 22: 5.6

Gambar 23: 5.7

Gambar 24: 6.1

```
admin@ MikroTik > /ip route print
Flags: X - disabled, A - active, D - dynamic
       S - static, F - floating, B - bgp, O - ospf, o - ovpn,
       B - blackhole, U - unreachable, P - prohibit
# DST ADDRESS      PRECIP-SRC      GATEWAY      DISTANCE
0 ADS 0.0.0.0/0      10.0.12.1       10.4.19.1     1
1 ADS 10.0.12.0/24   10.0.12.1       ether1        0
2 ADS 10.0.12.0/24   10.0.12.1       ether1        0
3 ADS 10.0.12.0/24   10.0.12.1       ether1        0
```

Gambar 25: 6.2

```
admin@ MikroTik > /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
```

Gambar 26: 6.3

```
admin@ MikroTik > ping 8.8.8.8
PING 8.8.8.8: 56 data bytes
0 8.8.8.8 36 111 21ms
1 8.8.8.8 36 111 21ms
2 8.8.8.8 36 111 21ms
icmp_seq=3 packet=10000 min-rtt=21ms, avg-rtt=21ms, max-rtt=21ms
```

Gambar 27: 6.4

```
FortiGate-VN04-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.3 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.0 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=3 ttl=254 time=1.1 ms
^C
-- 10.0.12.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms
```

Gambar 28: 6.5.1

```
FortiGate-VN04-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.5 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=0.6 ms
^C
-- 10.0.12.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/3.7/8.6 ms
```

Gambar 29: 6.5.2

7. Tugas Modul 7

```
Switch(config) #show vlan
VLAN Name                Status    Ports
-----
1  default              active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
10 SLES                 active    Gi0/1
40 OPERATIONS           active    Gi0/2, Gi0/3
100 token-ring-default   act/unsup
104 token-ring-default   act/unsup
105 token-ring-default   act/unsup
105 token-ring-default   act/unsup
Switch#show vlan brief
Switch#
```

Gambar 30: 7.1

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gi0/0     on        802.1q         trunking      1
Gi0/0     Vlan allowed on trunk
30,40
Gi0/0     Vlan allowed and active in management domain
30,40
Gi0/0     Vlan in spanning tree forwarding state and not pruned
30,40
Switch#
```

Gambar 31: 7.2

```
admin@ MikroTik > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS      NETWORK      INTERFACE
0 192.168.30.1/24 192.168.30.0 vlan30-sales
1 192.168.40.1/24 192.168.40.0 vlan40-operations
2 10.10.200.2/20  10.10.200.0 ether1
```

Gambar 32: 7.3

```
admin@ MikroTik > /ip dhcp-server print
Flags: D - dynamic, X - disabled, I - invalid
# NAME      INTERFACE  RELAY      ADDRESS POOL  LEASE TIME ADD-ADD
0 dhcp30   vlan30-sales  pool30     10s
```

Gambar 33: 7.4

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

Gambar 34: 7.5

#	VST ADDRESS	PREF-SEC	GATEWAY	DISTANCE
0	8.8.8.8/8		10.10.200.1	1
1	10.10.200.0/24		eth0/1	0
2	192.168.30.0/24		vlan30-sales	0
3	192.168.40.0/24		vlan40-operations	0

Gambar 35: 7.6

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

Gambar 36: 7.7

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

Gambar 37: 7.8

8. Tugas modul 8

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

Gambar 38: 8.1

#	Name	Range
0	port30	192.168.30.180-192.168.30.200

Gambar 39: 8.2

```

config firewall policy
edit 1
set name "LAN-TO-INTERNET"
set uuid 153c4bc-6503-51f1-2b6b-03775fd08f42
set srcintf "port2"
set dstintf "port1"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set name "LAN-to-GRE"
set uuid 447f60e0-6509-51f1-25b6-83c7a93f26af
set srcintf "port2"
set dstintf "GRE-SBY-JKT"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "GRE-to-LAN"
set uuid 448c77d2-6509-51f1-a24f-2ca1b8f7fb82
set srcintf "GRE-SBY-JKT"
set dstintf "port2"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"

```

Gambar 40: 8.3

```

FortiGate-VN04-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=112 time=22.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=23.3 ms
^C
-- 8.8.8.8 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 22.0/22.8/23.3 ms

```

Gambar 41: 8.4

```

FortiGate-VN04-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=1.2 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.1 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.2 ms
^C
-- 172.16.0.1 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/1.1/1.2 ms

```

Gambar 42: 8.5

```

FortiGate-VN04-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State          Dead Time   Address        Interface
1.1.1.1         1  Full/ -        00:00:31    172.16.0.1     GRE-SBY-JKT

```

Gambar 43: 8.6

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing Table for VN04:
E2    192.168.16.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
E2    192.168.16.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
E2    192.168.16.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]

```

Gambar 44: 8.7

9. Tugas Modul 9

```

FortiGate-VN04-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.5 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=0.6 ms
^C
-- 10.0.12.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/3.7/8.6 ms

```

Gambar 45: 9.1.1

```

FortiGate-VN04-KVM # execute ping 10.0.13.2
PING 10.0.13.2 (10.0.13.2): 56 data bytes
64 bytes from 10.0.13.2: icmp_seq=0 ttl=254 time=1.3 ms
64 bytes from 10.0.13.2: icmp_seq=1 ttl=254 time=1.0 ms
64 bytes from 10.0.13.2: icmp_seq=2 ttl=254 time=1.1 ms
^C
-- 10.0.13.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms

```

Gambar 46: 9.1.2

```

FortiGate-VN04-KVM # execute ping 172.16.0.2
PING 172.16.0.2 (172.16.0.2): 56 data bytes
64 bytes from 172.16.0.2: icmp_seq=0 ttl=255 time=3.9 ms
64 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=255 time=1.2 ms
^C
-- 172.16.0.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.2/2.2/3.9 ms

```

Gambar 47: 9.2.1

```

FortiGate-VN04-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=1.4 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.4 ms
^C
--- 172.16.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.4/2.4/4.4 ms

```

Gambar 48: 9.2.2

```

FortiGate-VN04-KVM # get router info ospf neighbor
ospf process 0, vrf 0
Neighbor ID Pri State Dead Time Address Interface
9.2.2.2 1 Full/ - 00:00:37 172.16.0.2 GRE-KT-SB7 GRE-KT-SB7

```

Gambar 49: 9.3.1

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VN04
O E2 192.168.30.0/24 [110/10] via 172.16.0.2, GRE-KT-SB7, 01:34:16, [1/0]
O E2 192.168.40.0/24 [110/10] via 172.16.0.2, GRE-KT-SB7, 01:34:16, [1/0]

```

Gambar 50: 9.4.1

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VN04
O E2 192.168.30.0/24 [110/10] via 172.16.0.1, GRE-SB7-KT, 01:47:12, [1/0]
O E2 192.168.30.0/24 [110/10] via 172.16.0.1, GRE-SB7-KT, 01:47:12, [1/0]
O E2 192.168.40.0/24 [110/10] via 172.16.0.1, GRE-SB7-KT, 01:47:12, [1/0]

```

Gambar 51: 9.4.2

```

VPCS> ping 192.168.40.10
64 bytes from 192.168.40.10 icmp_seq=1 ttl=60 time=13.884 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=60 time=11.788 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=60 time=10.312 ms
^C
VPCS>

```

Gambar 52: 9.5

```

VPCS> ping 192.168.10.101
64 bytes from 192.168.10.101 icmp_seq=1 ttl=60 time=12.882 ms
64 bytes from 192.168.10.101 icmp_seq=2 ttl=60 time=7.288 ms
64 bytes from 192.168.10.101 icmp_seq=3 ttl=60 time=7.034 ms
^C
VPCS>

```

Gambar 53: 9.6

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VN04
O E2 192.168.30.0/24 [110/10] via 172.16.0.2, GRE-KT-SB7, 01:34:16, [1/0]
O E2 192.168.40.0/24 [110/10] via 172.16.0.2, GRE-KT-SB7, 01:34:16, [1/0]

```

Gambar 54: 9.7.1

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VN04
O E2 192.168.30.0/24 [110/10] via 172.16.0.1, GRE-SB7-KT, 01:47:12, [1/0]
O E2 192.168.30.0/24 [110/10] via 172.16.0.1, GRE-SB7-KT, 01:47:12, [1/0]
O E2 192.168.40.0/24 [110/10] via 172.16.0.1, GRE-SB7-KT, 01:47:12, [1/0]

```

Gambar 55: 9.7.2

10. Tugas modul 10

```

VPCS> ip dhcp
DORA IP 192.168.10.102/24 GW 192.168.10.1

```

Gambar 56: 10.1

```

VPCS> ip dhcp
DORA IP 192.168.30.200/24 GW 192.168.30.1

```

Gambar 57: 10.2

```

VPCS> ping 8.8.8.8
64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=28.083 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=25.015 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=28.061 ms
^C
VPCS>

```

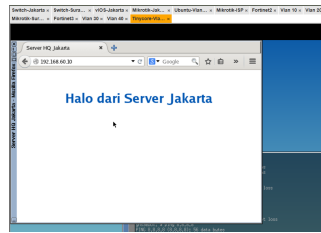
Gambar 58: 10.4

```

VPCS> ping 192.168.40.10
64 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.299 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=4.425 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.837 ms
VPCS>

```

Gambar 59: 10.5



Gambar 60: 10.6

```

fortigate-9964-vm # get router info ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
1.1.1.1 1 Full/- 00:00:31 172.16.0.1 GRE-SRV-JKT

```

Gambar 61: 10.7

```

VPCS> trace 192.168.40.10
Trace to 192.168.40.10, 8 hops max, press Ctrl-C to stop
 1 192.168.18.2 11.536 ms 2.351 ms 6.479 ms
 2 10.10.100.1 8.313 ms 3.027 ms 2.504 ms
 3 172.16.0.2 5.405 ms 3.748 ms 11.021 ms
 4 10.10.200.2 5.880 ms 5.079 ms 4.707 ms
 5 192.168.40.10 17.332 ms (100% type 3, code 1, destination port unreachable)

```

Gambar 62: 10.8

4 Kesimpulan

Berdasarkan praktikum yang telah dilakukan, dapat disimpulkan bahwa FortiGate berhasil dikonfigurasi sebagai perangkat keamanan jaringan sekaligus gateway utama yang menghubungkan jaringan internal dengan jaringan eksternal. Konfigurasi interface, routing, firewall policy, NAT, GRE Tunnel, dan OSPF dapat diterapkan dengan baik sehingga seluruh perangkat pada topologi jaringan mampu berkomunikasi sesuai dengan kebutuhan yang telah ditentukan pada modul praktikum.

Hasil pengujian menunjukkan bahwa konfigurasi routing statis dan routing dinamis menggunakan OSPF berhasil dilakukan. Jaringan internal Jakarta dapat terhubung dengan jaringan cabang Surabaya melalui GRE Tunnel yang berfungsi sebagai jalur komunikasi virtual antar lokasi. Selain itu, konfigurasi NAT dan firewall policy pada FortiGate memungkinkan perangkat dalam jaringan internal untuk mengakses internet melalui MikroTik ISP. Hasil tersebut sesuai dengan teori yang menjelaskan bahwa routing berfungsi menentukan jalur paket data, sedangkan NAT diperlukan untuk menerjemahkan alamat IP private agar dapat berkomunikasi dengan jaringan publik.

Selama proses praktikum ditemukan beberapa kendala, seperti belum adanya konfigurasi NAT pada MikroTik ISP, belum tersedianya route tertentu pada FortiGate, serta permasalahan DNS pada Ubuntu Server yang menyebabkan proses *apt update* gagal meskipun koneksi internet telah tersedia. Setelah dilakukan analisis dan perbaikan konfigurasi, seluruh layanan dapat berjalan dengan normal. Kondisi ini menunjukkan bahwa keberhasilan suatu jaringan tidak hanya bergantung pada satu perangkat saja, tetapi juga pada keterpaduan konfigurasi routing, NAT, firewall policy, dan DNS.

Melalui praktikum ini, praktikan memperoleh pemahaman yang lebih mendalam mengenai implementasi FortiGate dalam lingkungan jaringan enterprise. Praktikan juga memahami proses troubleshooting jaringan secara sistematis dengan melakukan pengujian konektivitas, analisis routing table, pengecekan firewall policy, verifikasi NAT, serta pengujian layanan DNS. Dengan demikian, tujuan

praktikum untuk memahami konfigurasi dan implementasi FortiGate sebagai perangkat keamanan dan penghubung jaringan dapat tercapai dengan baik sesuai dengan teori yang telah dipelajari.

5 Lampiran



Gambar 63: foto